ATOM-SG Baseline Assessment Test

Primary 6 Mathematics — Geometry Track

Name: _____ **Class:** _____ **Date:** _____

Instructions:

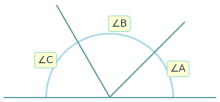
- This paper consists of 25 questions. Time allowed: 45 minutes.
- Show all your working clearly in the spaces provided.
- Marks may be awarded for correct method even if answer is wrong.
- Calculators are NOT allowed.
- Diagrams are NOT drawn to scale unless stated otherwise.

Q1. [A] [3m] The diagram shows angles ∠AOB, ∠BOC and ∠COD.

(a) Measure ∠AOB.
(b) Measure ∠BOC.
(c) Measure ∠COD.

G1: Measure angles using protractor

G001: Measure angles with protractor



Measure angles ∠A, ∠B, and ∠C in the diagram

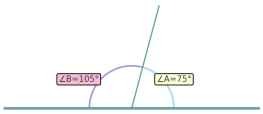
Working:

Ans: (a) _____° (b) _____° (c) _____°

Q2. [B] [2m] In the diagram, AOB is a straight line. ∠AOC = 75°.
Find ∠COB.

G1: Angles on a straight line

G002: Angles on straight line



Two angles on a straight line

Working:

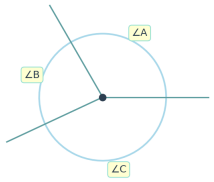
Ans: ∠COB = _____°

Baseline Assessment Test — Continued

Q3. [B] [2m] In the diagram, three angles meet at point O.
$\angle AOB = 120°$ and $\angle BOC = 85°$.
Find $\angle AOC$.

G1: Angles at a point

G003: Angles around a point



Three angles meet at a point

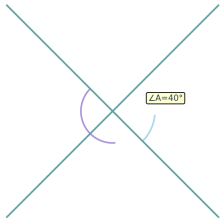
Working:

Ans: $\angle AOC$ = _____°

Q4. [B] [3m] In the diagram, straight lines AB and CD intersect at O.
$\angle AOC = 40°$.
(a) Find $\angle AOD$.
(b) Find $\angle BOD$.
(c) Find $\angle BOC$.

G1: Vertically opposite angles

G004: Vertically opposite angles



Two straight lines intersect

Working:

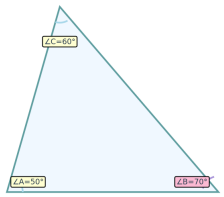
Ans: (a) _____° (b) _____° (c) _____°

Baseline Assessment Test — Continued

Q5. [C] [2m] In triangle ABC, ∠ABC = 50° and ∠BCA = 70°.
Find ∠CAB.

G1: Angle sum of triangle

G005: Triangle angle sum



Triangle ABC with two angles given

Working:

Ans: ∠CAB = _____°

Q6. [A] [2m] The diagram shows a figure with lines AB, CD and EF.
(a) Identify one pair of perpendicular lines.
(b) Identify one pair of parallel lines.

G1: Parallel and perpendicular lines

[Diagram: G006-parallel-perpendicular-v1]

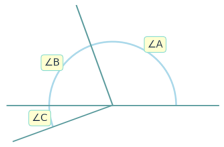
Working:

Ans: (a) _____ ⊥ _____ (b) _____ ∥ _____

Q7. [D] [3m] In the diagram, AOB is a straight line.
∠BOD = 90° and ∠AOC = 110°.
Find ∠COD.

G1: Combined angle properties

G007: Combined angle properties



Two-step angle reasoning

Working:

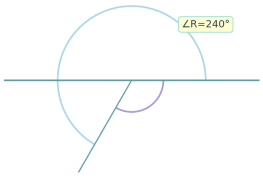
Ans: ∠COD = _____°

Baseline Assessment Test — Continued

Q8. [C] [2m] The diagram shows angle ∠XOY which is reflex.
Measure and write down the size of the reflex angle ∠XOY.

G1: Reflex angles

G008: Reflex angle measurement



Measure reflex angle ∠R

Working:

Ans: Reflex ∠XOY = _____°

Q9. [A] [2m] The diagram shows rectangle ABCD.
AB = 8 cm and BC = 5 cm.
Find the perimeter of rectangle ABCD.

G2: Perimeter of rectangle

G009: Rectangle perimeter

Working:

Ans: Perimeter = _____ cm

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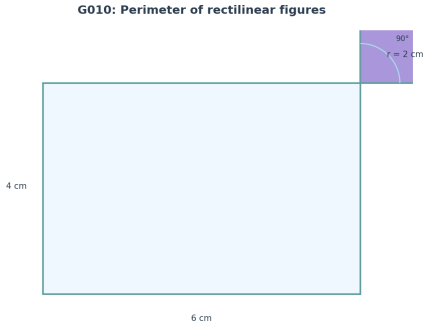
Q10. [B] [3m] The figure is made up of two rectangles.
Find the perimeter of the figure.

G2: Perimeter of composite figure



Working:

Ans: Perimeter = _____ cm

Q11. [B] [3m] The figure shows an L-shaped figure made up of two rectangles.
Find the area of the figure.

G2: Area of composite figure

[Diagram: G011-lshape-area-v1]

Working:

Ans: Area = _____ cm²

Q12. [C] [3m] A rectangular sheet of paper measures 12 cm by 8 cm.
A square of side 3 cm is cut from one corner as shown.
Find the area of the remaining sheet.

G2: Area with missing section

[Diagram: G012-cutout-rectangle-v1]

Working:

Ans: Area = _____ cm²

Q13. [A] [3m] Fill in the blanks.
(a) 3.5 m = _____ cm
(b) 250 cm = _____ m
(c) 4.2 km = _____ m

G2: Unit conversion (length)

Working:

Ans: (a) _____ (b) _____ (c) _____

Baseline Assessment Test — Continued

Q14. [B] [2m] Convert 2.5 m² to square centimetres.
2.5 m² = _____ cm²

G2: Unit conversion (area)

Working:

Ans: 2.5 m² = _____ cm²

Q15. [A] [2m] The diagram shows a regular hexagon.
(a) Draw all the lines of symmetry on the hexagon.
(b) How many lines of symmetry does a regular hexagon have?

G4: Symmetry

[Diagram: G015-hexagon-symmetry-v1]

Working:

Ans: (b) _____ lines

Q16. [C] [3m] The diagram shows half of a symmetrical figure.
The dotted line is the line of symmetry.
Complete the symmetrical figure.

G4: Complete symmetrical figure

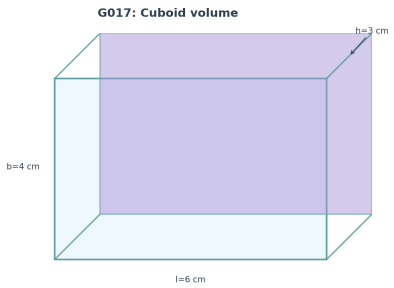
[Diagram: G016-symmetry-complete-v1]

Working:

Ans: (Draw on diagram)

Q17. [A] [2m] The diagram shows a cuboid.
Length = 6 cm, Breadth = 4 cm, Height = 3 cm.
Find the volume of the cuboid.

G3: Volume of cuboid

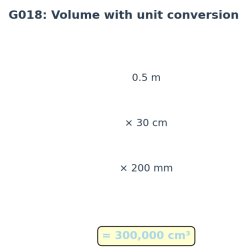


Working:

Ans: Volume = _____ cm³

Q18. [B] [3m] A box measures 0.5 m by 30 cm by 200 mm.
Find the volume of the box in cubic centimetres.

G3: Volume with unit conversion



Working:

Ans: Volume = _____ cm³

Q19. [C] [2m] Which of the following nets can be folded to form a cube?
Circle the letter(s) of the correct answer(s).

G3: Nets of cube

[Diagram: G019-cube-nets-v1]

Working:

Ans: Answer: _____

Q20. [D] [4m] The diagram shows a cuboid on an isometric grid.
Draw a net of the cuboid on the grid provided.

G3: Draw net of cuboid

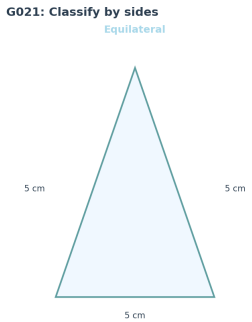
[Diagram: G020-cuboid-net-v1]

Working:

Ans: (Draw on grid)

Q21. [A] [1m] The diagram shows triangle ABC.
AB = 5 cm, BC = 5 cm and CA = 5 cm.
What type of triangle is ABC?

G4: Classify triangle by sides



Working:

Baseline Assessment Test — Continued

Q22. [A] [1m] Triangle PQR has angles 45°, 45° and 90°.
What type of triangle is PQR?

G4: Classify triangle by angles

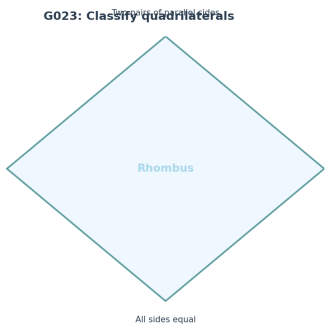
[Diagram: G022-triangle-angles-v1]

Working:

Ans: _____ triangle

Q23. [B] [2m] The diagram shows quadrilateral PQRS.
PQ ■ SR, PS ■ QR and PQ = QR = RS = SP.
However, ∠PQR ≠ 90°.
What type of quadrilateral is PQRS?

G4: Classify quadrilateral

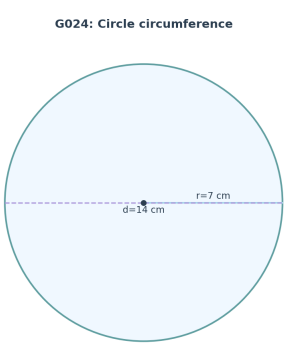


Working:

Ans: _____

Q24. [C] [2m] The diagram shows a circle with centre O.
The radius of the circle is 7 cm.
Find the circumference of the circle.
(Take π = 22/7)

G2: Circumference of circle

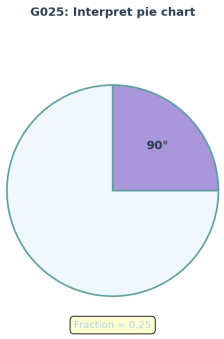


Working:

Ans: Circumference = _____ cm

Q25. [D] [3m] The pie chart shows how a student spends 24 hours in a day.
The angle of the 'Sleep' sector is 90°.
What fraction of the day does the student spend sleeping?
(Give your answer in the simplest form.)

G2: Pie chart as fraction



Working:

Ans: Fraction = _____

Answer Key — For Markers Only

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Q#	Answer	Subpathway	
Q 1	(a) _____° (b) _____° (c) _____°	G1: Measure angles using pro...	3m
Q 2	∠COB = _____°	G1: Angles on a straight lin...	2m
Q 3	∠AOC = _____°	G1: Angles at a point	2m
Q 4	(a) _____° (b) _____° (c) _____°	G1: Vertically opposite angl...	3m
Q 5	∠CAB = _____°	G1: Angle sum of triangle	2m
Q 6	(a) _____ ⊥ _____ (b) _____ ■ __...	G1: Parallel and perpendicul...	2m
Q 7	∠COD = _____°	G1: Combined angle propertie...	3m
Q 8	Reflex ∠XOY = _____°	G1: Reflex angles	2m
Q 9	Perimeter = _____ cm	G2: Perimeter of rectangle	2m
Q10	Perimeter = _____ cm	G2: Perimeter of composite f...	3m
Q11	Area = _____ cm²	G2: Area of composite figure	3m
Q12	Area = _____ cm²	G2: Area with missing sectio...	3m
Q13	(a) _____ (b) _____ (c) _____	G2: Unit conversion (length)	3m
Q14	2.5 m² = _____ cm²	G2: Unit conversion (area)	2m
Q15	(b) _____ lines	G4: Symmetry	2m
Q16	(Draw on diagram)	G4: Complete symmetrical fig...	3m
Q17	Volume = _____ cm³	G3: Volume of cuboid	2m
Q18	Volume = _____ cm³	G3: Volume with unit convers...	3m
Q19	Answer: _____	G3: Nets of cube	2m
Q20	(Draw on grid)	G3: Draw net of cuboid	4m
Q21	_____ triangle	G4: Classify triangle by sid...	1m
Q22	_____ triangle	G4: Classify triangle by ang...	1m
Q23	_____	G4: Classify quadrilateral	2m
Q24	Circumference = _____ cm	G2: Circumference of circle	2m
Q25	Fraction = _____	G2: Pie chart as fraction	3m